

Hawkeye EQMS · Risk Register (FMEA) · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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Live frontend	https://hawkeye-frontend-dev-chi.vercel.app
Live backend	https://hawkeye-backend-dev.vercel.app

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1. Module overview

Module	Risk Register (FMEA)
Pharma purpose	FMEA-style hazard register. Capture process-step risks (failure mode → effect → cause), score with Severity × Occurrence × Detectability = RPN, add mitigations until residual risk acceptable.
Compliance basis	ISO 9001:2015 §6.1 (risk-based thinking) · ICH Q9 (R1) Quality Risk Management · ISO 14971 (med device risk mgmt)
Hawkeye routes	/risk-register (frontend) · /api/risk-items (backend)
Hawkeye model file	backend/src/models/RiskItemModel.js
Tenant module flag	modules.RISK_MANAGEMENT

Each risk row captures S/O/D scores (1-10) which auto-compute RPN = S × O × D. RPN bands: <60 LOW, 60-124 MEDIUM, 125-199 HIGH, ≥200 CRITICAL. Risks move OPEN → MITIGATED (after at least one mitigation completed and residual S/O/D entered) → ACCEPTED → CLOSED. An optional Wave-2 AI brainstormer pre-suggests failure modes / effects / causes from a process description.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. MET = parity, PARTIAL = ships but with caveats, GAP = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
FMEA scoring (S × O × D) ICH Q9	severity / occurrence / detectability fields (1-10 each); RPN auto-computed in pre-save hook.	MET	
Risk band classification (LOW / MED / HIGH / CRITICAL) ICH Q9	riskBand auto-set from RPN thresholds: <60 LOW, 60-124 MED, 125-199 HIGH, ≥200 CRITICAL.	MET	
Residual risk after mitigation ICH Q9 Annex II	residualSeverity / residualOccurrence / residualDetectability fields; residualRpn auto-computed when all three set. Pre-save hook fires (fixed in v2.1).	MET	
Mitigation actions with owner + due date + status ICH Q9 §6	mitigations[] subdocument array. Each: action / owner / dueDate / status (OPEN/IN_PROGRESS/COMPLETED/OVERDUE) / residualRpn.	MET	
Risk source tracking (audit / deviation / complaint / CAPA / inspection) ICH Q10 §3.2.5	sourceType enum: AUDIT_FINDING / DEVIATION / COMPLAINT / MANUAL / CAPA / INSPECTION / REGULATORY.	MET	
Risk categories (Quality / Safety / Regulatory / Operational / Environmental / Financial) —	riskCategory enum with 7 values (incl. CUSTOM).	MET	
Lifecycle: OPEN → MITIGATED → ACCEPTED → CLOSED + TRANSFERRED ISO 9001 §6.1	5-state enum. PUT /:id sets status; POST /:id/mitigate appends mitigation.	MET	
AI brainstorm of failure modes from process description FDA AI guidance Jan 2025	Wave 2 endpoint POST /api/ai/risk/brainstorm-scenarios — given processDescription + productClass returns failure modes / effects / causes.	MET	
Risk owner identification + identifiedBy traceability ISO 9001 §6.1.2	riskOwner + identifiedBy ObjectId refs to users; both required.	MET	
Approver sign-off on ACCEPTED state (residual risk acceptable) ICH Q9 §5.4	Status change to ACCEPTED currently allowed by tenant_admin role; no e-signature gate.	PARTIAL	Wire e-sig requirement on ACCEPTED + CLOSED
Residual-risk scheduler (re-check on cadence) ISO 9001 §6.1.2 (g)	No nextReviewDate field today.	GAP	Add nextReviewDate + reminder on threshold crossings

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026`. Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()`.

Persona	Email	Responsibilities in this module	Lifecycle states they touch
QA Specialist Kenji admin · risk identifier	qa.specialist@novex-pharma.demo	Logs new risks; assigns mitigation actions; updates residual scores when mitigations complete.	OPEN MITIGATED
Head of QA James admin · risk owner	qa.head@novex-pharma.demo	Owns the register, reviews RPN movements, escalates CRITICAL band to VP.	OPEN MITIGATED
VP Quality Elena tenant_admin · approver	vp.quality@novex-pharma.demo	Approves residual risk acceptance + closure.	ACCEPTED CLOSED TRANSFERRED
Lars Nilsson Maintenance Eng (mitigation owner)	maintenance@novex-pharma.demo	Owns equipment-related mitigations.	OPEN (mitigation execution)

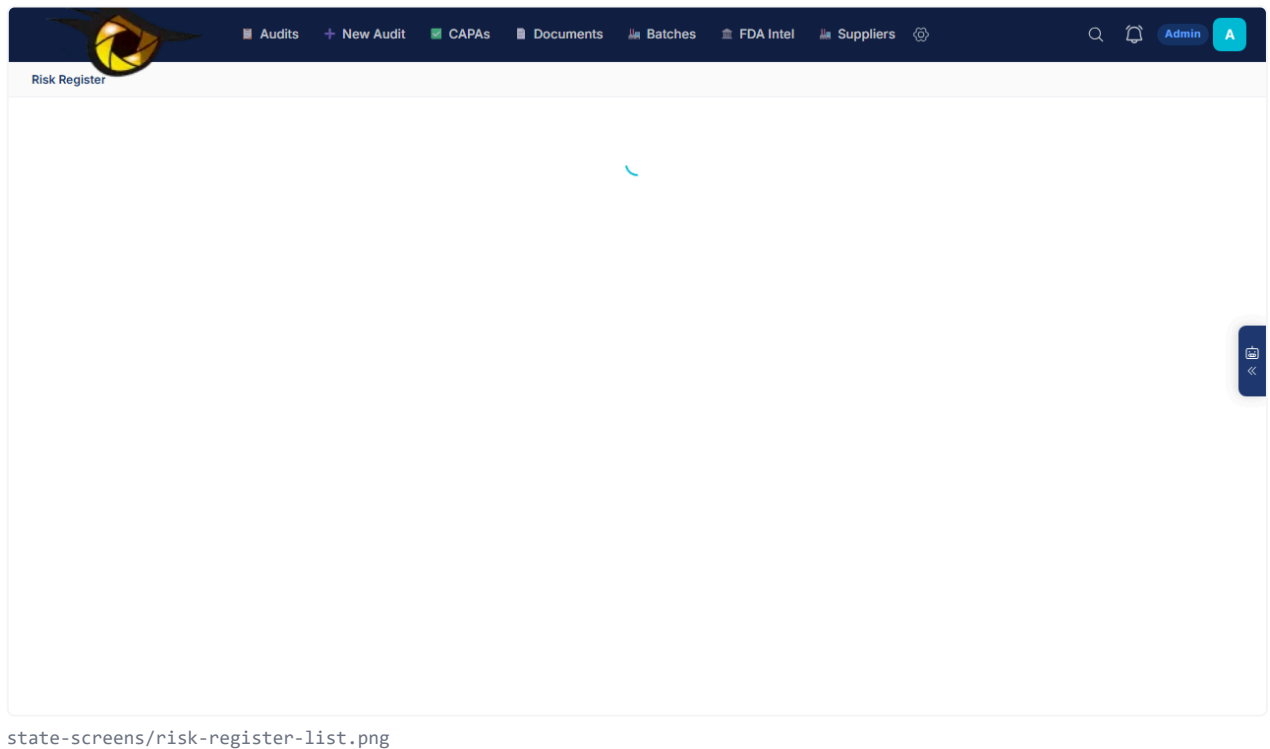
4. Feature catalogue (click-by-click)

4.1 Risk register

What it does	Tabular FMEA register: process step / failure mode / effect / S / O / D / RPN / band / status / mitigations count.
Menu / route	/risk-register
Available to roles	any tenant viewer
Backend API	GET /api/risk-items

CLICK-BY-CLICK

- NAVIGATE** Click 'Risk' in the top nav
→ Page loads with the 5 seeded risks
- WAIT** Spinner clears
→ 5 rows: Blending RPN=240 CRITICAL · EM excursion 189 HIGH · Material misship 160 HIGH · Compression 140 HIGH · Stability 96 MEDIUM



4.2 Filter by Risk Band / Status

What it does	Drop-downs above the table.
Menu / route	<code>/risk-register · top filter row</code>
Available to roles	<code>any viewer</code>

CLICK-BY-CLICK

1. **CLICK** Click 'Risk Band' dropdown
→ LOW / MEDIUM / HIGH / CRITICAL
2. **CLICK** Click 'Status' dropdown
→ OPEN / MITIGATED / ACCEPTED / CLOSED / TRANSFERRED

4.3 + Log Risk dialog

What it does	Capture a new FMEA risk row. RPN computes automatically.
Menu / route	<code>/risk-register · top-right '+ Log Risk' button</code>
Available to roles	<code>any tenant user</code>
Backend API	<code>POST /api/risk-items</code>

CLICK-BY-CLICK

1. **CLICK** Click '+ Log Risk'
→ Dialog opens with FMEA form
2. **TYPE** Fill processStep (e.g. 'Blending · Line 2')
→ required

3. **TYPE** Fill failureMode (one sentence)
→ required
4. **TYPE** Fill failureEffect (consequence)
→ required
5. **TYPE** (Optional) Fill failureCause
6. **CLICK** Pick severity (1-10)
→ required; 10 = catastrophic patient harm
7. **CLICK** Pick occurrence (1-10)
→ required; 10 = highly likely
8. **CLICK** Pick detectability (1-10)
→ required; 10 = nearly impossible to detect
9. **CLICK** Pick riskCategory (QUALITY / SAFETY / REGULATORY / OPERATIONAL / ENVIRONMENTAL / FINANCIAL / CUSTOM)
→ default QUALITY
10. **CLICK** Click 'Save'
→ Row appears with RPN auto-computed; band auto-set

FIELDS CAPTURED

Field	Required	Validation / values	Notes
processStep	Yes	string	
failureMode	Yes	string	
failureEffect	Yes	string	
severity	Yes	1-10	
occurrence	Yes	1-10	
detectability	Yes	1-10	
rpn	No	auto-computed (S × O × D)	pre-save hook
riskBand	No	auto-set from RPN	<60 LOW, 60-124 MED, 125-199 HIGH, ≥200 CRITICAL
identifiedBy	Yes	ObjectId	ref to user
riskOwner	Yes	ObjectId	ref to user

4.4 AI · Brainstorm risk scenarios (Wave 2)

What it does	Given a process description + product class, AI proposes 5-10 failure modes with effects + causes you can edit + save as risk-items.
Menu / route	(API today) POST /api/ai/risk/brainstorm-scenarios – UI button on roadmap
Available to roles	admin tenant_admin
Backend API	POST /api/ai/risk/brainstorm-scenarios
AI assistance	AI Free Gemini 2.5 Flash-Lite; output schema: failure_modes[] each with mode/effect/cause/suggestedSOD

CLICK-BY-CLICK

1. **API** POST with { processName, processDescription, productClass, equipmentInvolved }
→ Returns 5-10 candidate failure modes with suggested S/O/D scores

4.5 + Add mitigation

What it does	Append a mitigation action with owner + dueDate. Stored as a subdocument under mitigations[].
Menu / route	Risk row · 'Mitigations N actions' link → drawer
Available to roles	any tenant user
Backend API	POST /api/risk-items/:id/mitigate

CLICK-BY-CLICK

1. **CLICK** Click the 'N actions' link on a risk row
→ Drawer with current mitigations + Add button
2. **CLICK** Click '+ Add Mitigation'
→ Form opens
3. **TYPE** Fill action description
→ required
4. **CLICK** Pick owner (user dropdown)
→ ObjectId
5. **CLICK** Pick dueDate
→ required
6. **CLICK** Click 'Save'
→ Mitigation appears with status=OPEN

4.6 Update risk (PUT)

What it does	Update any field including residual S/O/D + status. PUT now triggers pre-save hook (fixed in v2.1).
Menu / route	Risk detail · 'Edit' button (or PUT API)
Available to roles	any tenant user
Backend API	PUT /api/risk-items/:id

CLICK-BY-CLICK

1. **API** PUT body: { status: 'MITIGATED', residualSeverity: 8, residualOccurrence: 2, residualDetectability: 3 }
→ residualRpn auto-computed = 48; status flips to MITIGATED

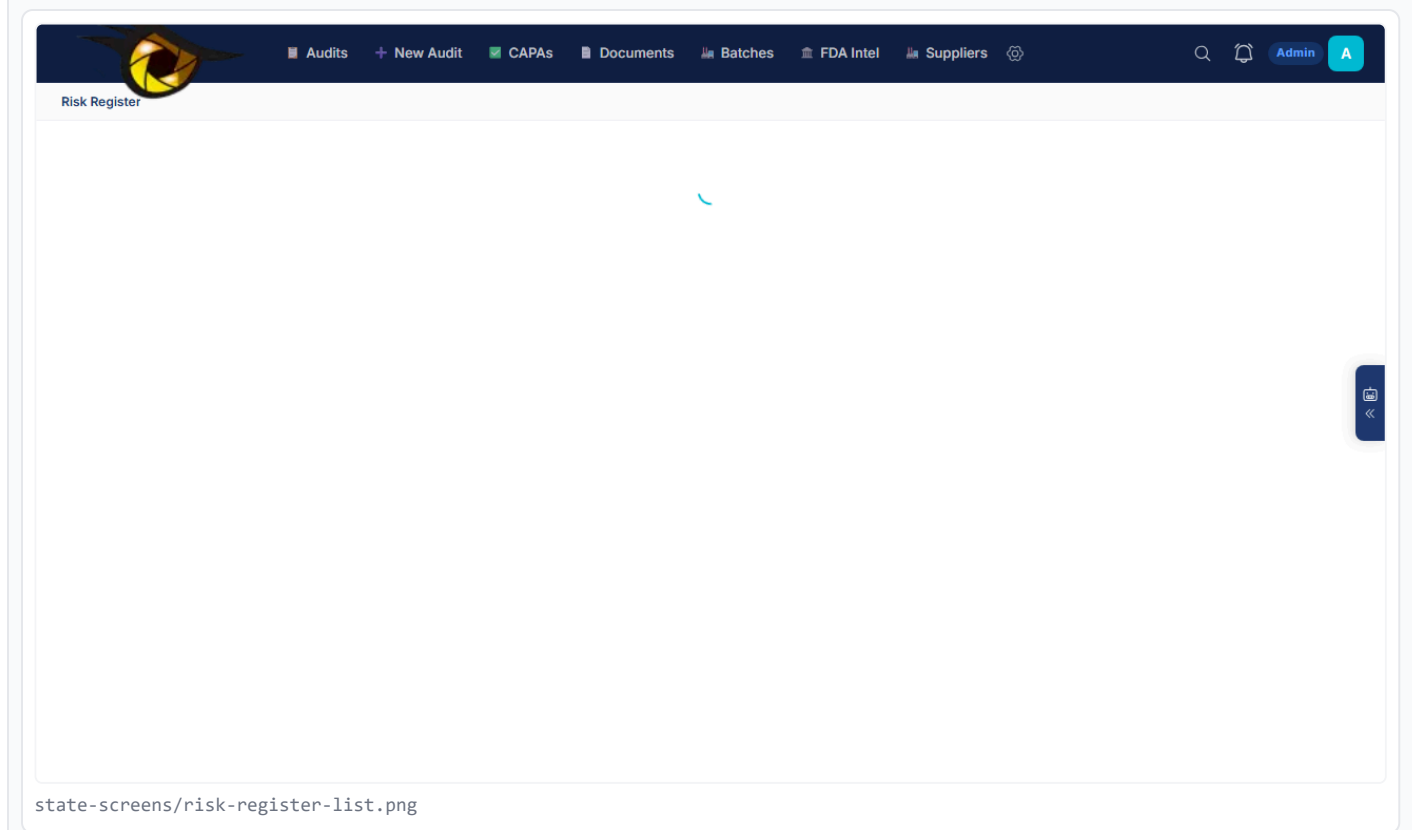
5. Full lifecycle walkthrough — one transaction, end-to-end

One risk walked from creation through mitigation, residual scoring, acceptance, and closure. Personas Kenji (identify) → Lars (mitigate) → Elena (accept + close).

#	Persona	State transition	Action	API	Observed	Result
1	Kenji QA Specialist	– → OPEN	Click '+ Log Risk' → fill FMEA fields with S=8, O=4, D=6 → Save	POST /api/risk-items	rpn=192 · band=HIGH · status=OPEN	PASS

Expected DB state after this step:

```
risk-items { processStep, failureMode, failureEffect, severity: 8, occurrence: 4, detectability: 6, rpn: 192, riskBand: 'HIGH', status: 'OPEN', identifiedBy: kenji._id, riskOwner: kenji._id }
```



2	Kenji QA Specialist	OPEN → OPEN (with mitigation)	Click 'Mitigations' on the row → '+ Add Mitigation' → fill action + owner=Lars + dueDate → Save	POST /api/risk-items/:id/mitigate	mitigations[] now has 1 entry status=OPEN	PASS
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Expected DB state after this step:

```
risk-items.mitigations += { action, owner: lars._id, dueDate, status: 'OPEN' }
```

3	Lars Maintenance (mitigation owner)	OPEN → OPEN (mitigation IN_PROGRESS)	Lars updates the mitigation: status=IN_PROGRESS	PUT /api/risk-items/:id (with updated mitigations[])	mitigations[0].status='IN_PROGRESS'	PASS
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#	Persona	State transition	Action	API	Observed	Result
4	Lars Maintenance	OPEN → MITIGATED	Mitigation complete. PUT with status=MITIGATED + residualSeverity:8 / residualOccurrence:2 / residualDetectability:3	PUT /api/risk-items/:id	Pre-save hook fires: residualRpn=48 (down from 192). status=MITIGATED. Backend bug fixed in v2.1 (load+save instead of findOneAndUpdate).	PASS
Expected DB state after this step: <pre>risk-items { status: 'MITIGATED', residualSeverity: 8, residualOccurrence: 2, residualDetectability: 3, residualRpn: 48 }</pre>						
5	Elena VP Quality	MITIGATED → ACCEPTED	PUT with status=ACCEPTED	PUT /api/risk-items/:id	status=ACCEPTED — residual risk signed off	PASS
Expected DB state after this step: <pre>risk-items.status = 'ACCEPTED'</pre>						
6	Elena VP Quality	ACCEPTED → CLOSED	PUT with status=CLOSED	PUT /api/risk-items/:id	status=CLOSED · terminal	PASS

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
Risk Scenario Brainstormer (Wave 2)	OPEN (creation)	POST /api/ai/risk/brainstorm-scenarios	(API only today; UI button on roadmap)	Given a process description, returns 5-10 candidate failure modes + effects + causes + suggested S/O/D	Free Gemini 2.5 Flash-Lite
(roadmap) Residual-risk re-evaluation prompt	MITIGATED	(future)	Auto-fired when nextReviewDate hits	Reminds owner to re-score S/O/D and confirm ACCEPTED state still valid	rule + email

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
OPEN	ISO 9001 §6.1.1 ICH Q9 §IV.1	processStep + failureMode + failureEffect + S/O/D + identifiedBy + riskOwner	risk-items
OPEN (with mitigations)	ICH Q9 §IV.2 ISO 9001 §6.1.2(c)	mitigations[] each with action + owner + dueDate + status	risk-items.mitigations
MITIGATED	ICH Q9 Annex II	residualSeverity + residualOccurrence + residualDetectability + residualRpn (auto-computed)	risk-items
ACCEPTED	ICH Q9 §5.4	Approver decision + acceptance rationale + (e-sig when wired)	risk-items + (audit trail)
CLOSED	ISO 9001 §6.1.2(g)	Final closure + closedBy	risk-items + auditTrail

8. Test results matrix

Suite	Scope	Result	Evidence
eqms-lifecycle.spec.ts · risk-item	Create → mitigate → MITIGATED with residual → ACCEPTED → CLOSED	PASS	6/6 PASS · eqms-test-results-v2.pdf
novex-walkthrough.spec.ts	/risk-register UI render with 5 seeded rows + RPN/Band columns	PASS	05-kenji.png + 19-elena.png
Backend bug fix verification	PUT /api/risk-items/:id triggers pre-save hook → residualRpn computed	PASS	v2.1 commit cf0de50

9. Known gaps + roadmap

- **Wire AI Brainstormer button into the +Log Risk dialog** — Today the API works but no UI button. Add a 'Suggest failure modes with AI' button next to processStep input. **HIGH**
- **Mandatory e-sig on ACCEPTED + CLOSED** — Wire /api/electronic-signatures/sign for residual-risk acceptance. **HIGH**
- **nextReviewDate scheduler** — Field + cron + reminder for periodic re-evaluation of MITIGATED risks. **MEDIUM**
- **Heatmap view (Severity × Occurrence)** — Visual grid where each cell is a risk. Drill-down on click. **MEDIUM**
- **Cross-link to source records (deviation/CAPA/audit)** — Today sourceType + sourceRefId are captured; surface them as clickable chips. **LOW**